DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTECPC401 - PYTHON PROGRAMMING

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.Use of non-programmable scientific calculators is allowed.
5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 In which year was Python first released?
a) 1989 b) 1991 c) 1994 d) 2000

2 Which of the following is a Python keyword?
a) define b) class c) variable d) print

3 What command checks Python version?
a) --version b) -v c) --check d) version()

4 Which sequence type is immutable?
a) List b) Tuple c) Set d) Dictionary

5 Which operator checks if an element is in a sequence?
a) is b) in c) and d) or

6 What keyword is used for multiple conditions in Python?
a) if b) elif c) else d) switch

7 What does a lambda function lack?
a) Arguments b) Name c) Return d) Scope

8 What is the scope of a variable defined inside a function?
a) Global b) Local c) Module d) Package

9 What is a namespace in Python?
a) Module b) Scope c) Package d) Function

10 What is encapsulation?
a) Inheriting b) Hiding c) Overriding d) Overloading

11 What is used to execute SQL queries in Python?
a) Cursor b) Connection c) Commit d) Fetch

12 Which module is used for data compression in Python?
a) zipfile b) os c) sys d) math

Q2. Solve the following.

A) Describe the role and significance of Python's building blocks: keywords, identifiers, variables, and indentation. Provide examples for each. 6

B) Write a Python program to concatenate two strings provided by the user and display the result in uppercase. 6

Q3. Solve the following.

A) Write a Python program to find the largest among three numbers entered by the user using nested if-else statements. 6

B) Discuss the various types of operators in. Explain operator precedence and associativity with examples. 6

Q4. Solve Any Two of the following.

A) Describe default and positional arguments in Python. Provide a code example of a function using both argument types. 6

B) Explain the role of PIP in Python package management. Provide the commands to install, uninstall, and list packages using PIP with explanations. 6

C) Write a Python program to define a recursive function that calculates the factorial of a number and handles negative input using a base case. 6

Q5. Solve Any Two of the following.

A) Describe encapsulation and data hiding in Python. Provide a code example to show how private attributes are implemented and accessed. 6

B) Write a Python program to create a class representing a rectangle with methods to calculate area and perimeter. 6

C) Explain the difference between errors and exceptions in Python. Provide examples of a syntax error and a runtime exception with explanations. 6

Q6. Solve Any Two of the following.

A) Write a Python program to write user-entered student details (name, roll number) to a CSV file and display its contents. 6

B) Write a Python program to draw a square spiral pattern using turtle graphics based on a user-entered number of iterations. 6

C) Discuss the datetime module in Python for handling date and time. Provide examples of formatting dates and calculating time differences. 6

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